Safaa Dabagh

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Teaching Philosophy

Statistics is one of the most powerful tools a citizen can hold, and mathematics education belongs to every student, not to a few. My practice rests on three commitments: access (free, advanced materials and a free statistical app, so no student is held back by cost), representation (open materials that show students who does statistics), and students as creators of knowledge. I see the classroom as the place where students discover themselves as leaders. Full statement: learnwithoutwalls.com/about.html

Education

University of California, Los Angeles (UCLA)
Master of Science in Statistics, June 2016 GPA: 3.83
Thesis: "An Analysis of the Discrepancy between the Students' Responses to Multiple-Choice and Open-Ended Questions"
Advisor: Professor Mahtash Esfandiari

Bachelor of Science in Mathematics/Economics, March 2010 GPA: 3.93
Minor: Statistics
Honors: Summa Cum Laude (2010), Dean's List (2007 to 2010)

West Los Angeles College
Associate of Arts in Liberal Arts and Science, June 2007 GPA: 4.00
Honors: Dean's and President's List (2005 to 2007)

Lebanese University, Tripoli
Certificate of Completion in Psychology, March 2003 GPA: 3.48

Teaching Experience

Loyola Marymount University, Los Angeles, CA Spring 2026 to Present
Instructor, Mathematics (MATH 205: Applied Statistics)

- Teach two sections of applied statistics with R; design all lectures, labs, theory sets and exams
- Lead students to compute and interpret in pairs, with analysis based exams

Santa Monica College, Santa Monica, CA Summer 2016 to Present
Mathematics Instructor (STAT C1000: Introduction to Statistics; MATH 54C: corequisite support; previously Math 4/4C: College Algebra for STEM)

- Teach evening statistics sections, including a section paired with its corequisite support course
- Created and taught Statistics for Journalism, built around real world data
- Moved all labs from paid software to free tools so every student has access

West Los Angeles College, Culver City, CA 2017 to Present
Mathematics Instructor (STAT C1000E for the LAPD POPP cadet program; Math 125, 227, 230, 236)

- Teach the statistics lecture and lab for the Police Orientation and Preparation Program cohort at the LAPD Ahmanson Recruit Training Center
- Earlier courses: Intermediate Algebra, Liberal Arts Math, Calculus for Business, and Statistics
- Mentor first-generation college students at the college where my own college education began

Joyce University of Nursing and Health Sciences (Online) Summer 2025
Statistics Instructor (MAT 300: Introduction to Statistics)

- Designed and taught a fully online statistics course for nursing and health sciences students
- Wrote the course text, *Statistics in the Eye of a Nurse*, with lectures, worksheets, exams and study guides

Los Angeles Mission College, Sylmar, CA Fall 2022
Mathematics Instructor (Math 115, 125, 227)

- Taught dual enrollment Elementary Algebra, Intermediate Algebra, and Introduction to Statistics
- Supported high school students moving into college level mathematics

UCLA Department of Statistics, Los Angeles, CA 2013 to 2016
Teaching Assistant (STATS 10, 13, 20, 112, 100A, 100C, 202AC); Teacher Associate Consultant (STAT 495: Teaching College Statistics, Winter 2014)

- Led discussion sections for undergraduate and graduate statistics courses and supported students in R
- Trained future teaching assistants in the practice and theory of teaching statistics

Open Educational Resources and Equity Work

You're Smarter Than You Think: Statistics 2026
Author | open textbook for neurodivergent learners, with R, jamovi, TI-84 and StatCrunch companions | CC BY 4.0
doi.org/10.5281/zenodo.21254374; listed in the Open Textbook Library

You're Smarter Than You Think: College Algebra for STEM Majors
Author | open textbook for neurodivergent students | CC BY

Stats Without Walls 2026
Creator | learnwithoutwalls.com/stats

- Free, browser based statistical software for introductory and applied statistics, built so no student pays for software
- Covers descriptive statistics, graphs, all introductory inference, ANOVA, chi-square, regression, non-parametric and advanced models; results checked against R

Learn Without Walls Founder
learnwithoutwalls.com: free, neurodivergent friendly courses, open textbooks, and an interactive statistics learning platform with adaptive scaffolding at four levels of support

ASCCC OERI Statistics for Social Justice 2020
Contributor, West Los Angeles College team | Canvas Commons

- Co-developed an open Canvas workbook for introductory statistics with a social justice focus, funded by the Academic Senate for California Community Colleges Open Educational Resources Initiative (CC BY-NC-SA)
- Includes materials on African American and Latinx mathematicians and on who does statistics

Statistics for Journalism, Santa Monica College
Course creator and instructor

- Brought mass shootings into the classroom as a real world example; students helped shape a state level analysis and proposed comparing rates per person
- The class data, reused in later courses, became the starting point of a research paper on economic disadvantage and mass shootings

Research and Publications

Dabagh, S. (2026). AI-assisted assessment design in mathematics and statistics: A multi-course Bloom's taxonomy analysis of cognitive demand shift. Revise and resubmit, *Teaching Statistics*.

Dabagh, S. (2026). Raising the floor, holding the bar: Distributional effects of AI-assisted course design in two community college mathematics courses. Preprint, EdArXiv.

Dabagh, S. (2026). Economic disadvantage and mass shootings across U.S. states, 2018 to 2024: A state-level panel analysis. Preprint, Research Square; submitted for journal review.

Dabagh, S. (2026). Rising success, persistent gaps: Student-level statistics outcomes across California's community college placement reform era. Manuscript in revision.

Dabagh, S. (in preparation). AI-assisted course design and student outcomes in community college mathematics: A two-course quasi-experimental comparison.

Dabagh, S. M. (2016). *An analysis of the discrepancy between the students' responses to multiple-choice and open-ended questions* (Master's thesis, University of California, Los Angeles). escholarship.org/uc/item/8dr9q96b

All papers and preprints: learnwithoutwalls.com/publications.html

Mobilize Project, Los Angeles, CA 2013 to 2014
Graduate Student Researcher (Principal Investigator: Rob Gould)

- Developed curriculum for LAUSD public schools, including Introduction to Data Science courses
- Trained teachers on statistical concepts, inquiry based teaching, and statistical software

Professional Development and Service

Open for Antiracism (OFAR) Program, Santa Monica College team 2026 to 2027
Year-long program on antiracist teaching and open pedagogy; action plan for introductory statistics

Google AI for Educators Certification 2025
Training in artificial intelligence applications for education

AI in Education Coursework 2024 to 2025
Coursera and FutureLearn: learning analytics and educational data mining, cognitive scaffolding, adaptive learning systems design, AI in educational technology

American Statistical Association (ASA) 2024 to Present
Member

Technical Skills

Programming Languages: R, Python, C++, Matlab, SAS, SPSS
Statistical Software: Advanced R (ggplot2, tidyverse, statistical modeling), jamovi, StatCrunch, TI-84
Web Development: HTML, CSS, JavaScript, GitHub Pages
Document Preparation: LaTeX, Markdown, Microsoft Office Suite
Educational Technology: Canvas, Brightspace, Blackboard, browser based instructional tools
Data Science: Statistical modeling, data visualization, machine learning, causal inference, sensitivity analysis
AI: Integrating large language models into teaching tools, prompt design, critical AI literacy

Pedagogical Expertise

Instructional Design: Curriculum development, assessment design, open educational resources, universal design for learning
Teaching Modalities: In person, hybrid, fully online (synchronous and asynchronous)
Methods: Active learning, inquiry based teaching, real world data projects, open pedagogy, adaptive scaffolding
Student Populations: First-generation students, community college students, neurodivergent learners, dual enrollment students, police cadets, health sciences students

Languages

Fluent in Arabic and English; conversational in French

Research Interests

Statistics education for all learners, AI in teaching and assessment design, equity in community college placement reform, open educational resources and open pedagogy, neurodivergent friendly design, applied statistics for public policy